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Inventor(s)	Umeda; Keisuke

Storage device for vehicle and method of assembling storage device

Abstract

A vehicle storage device includes a storage portion that has an opening in its one face and is configured to store at least one thing therein, a bracket body fixed to an attaching surface portion, which is another face of the storage portion, a shaft member supported on the bracket body, a lid supported on the shaft member such that the lid can swing to open and close at least part of the opening of the storage portion, and a coil spring configured to bias the lid in an opening direction. The bracket body includes at least one engaging portion that extends along the attaching surface portion. The attaching surface portion includes at least one engaged portion that has an opening to receive the respective engaging portion.

Inventors:	Umeda; Keisuke (Tokyo, JP)
Applicant:	MORIROKU TECHNOLOGY COMPANY, LTD. (Tokyo, JP)
Family ID:	1000008777612
Assignee:	MORIROKU CO., LTD. (Tokyo, JP)
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Primary Examiner: Morrow; Jason S

Attorney, Agent or Firm: Rankin, Hill & Clark LLP

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a technology for assembling a storage device for a vehicle.

BACKGROUND OF THE INVENTION

(2) For example, a storage device for a vehicle is provided between a driver's seat and a passenger's seat in a passenger vehicle for receiving and storing goods and belongings (i.e., things) of a person in the vehicle. One of the conventional storage devices for vehicles is disclosed in JP-A-2012-1103.

(3) The vehicle storage device disclosed in JP-A-2012-1103 includes a box body having an open upper surface, a shaft-supporting bracket fixed on a rear surface of the box body for supporting a rotating shaft, a swingable lid body supported by the shaft-supporting bracket via a movable bracket such that the lid body can open and close, and a torsion coil spring disposed on the shaft-supporting bracket for biasing the lid body in the opening direction via the movable bracket.

SUMMARY OF THE INVENTION

(4) An assembly work of such a vehicle storage device may be carried out by, for example, temporarily assembling and uniting the brackets, a shaft member, the coil spring and the lid body and placing the brackets in predetermined positions of a storage portion.

(5) However, in many cases, the lid is large and heavy as compared to the brackets. Therefore, when placing the brackets in the predetermined positions of the storage portion, it is necessary to perform the placing work while supporting the weight of the lid. Alternatively, the lid may be closed and held by a holding element provided at the storage portion, and then the brackets may be placed in the predetermined positions of the storage portion. In this case, however, it is necessary to perform the placing work while pressing the brackets toward the storage portion against the biasing force of the coil spring.

(6) On the other hand, if the assembly work is carried out by placing the brackets in the predetermined positions of the storage portion and subsequently attaching the coil spring and other members, it is difficult to secure the space required for the work. Therefore, this assembly work cannot be employed.

(7) An object of the present invention is to provide a storage device for a vehicle which can be easily assembled.

(8) According to one aspect of the present invention, there is provided a storage device for a vehicle that includes: a storage portion that has an opening in its one face and is configured to store thing therein; a bracket body fixed to an attaching surface portion, which is another face of the storage portion; a shaft member supported on the bracket body; a lid supported on the shaft member such that the lid can swing to open and close at least part of the opening of the storage portion; and a coil spring configured to bias the lid in an opening direction, wherein the bracket body includes at least one engaging portion that extends along the attaching surface portion, and the attaching surface portion includes at least one engaged portion that has an opening to receive the respective engaging portion.

(9) According to another aspect of the present invention, there is provided a method of assembling a storage device for a vehicle. The method includes: a preparation step of preparing a storage portion that has one opening in its one face and at least one engagement portion formed on an attaching surface of the storage portion, a bracket body that has at least one engaging portion, a shaft member, a lid, and a coil spring; a temporary assembly step of temporarily assembling the bracket body, the shaft member, the lid, and the coil spring; a temporary fixing step of inserting the respective engaging portion of the bracket body, on which the lid and the coil spring are temporarily assembled, into the respective engaged portion to temporarily fix the bracket body to the storage portion; and a fixing step of fixing the bracket body, which is temporarily fixed to the storage portion, to the attaching surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a vehicle storage device according to an embodiment of the

invention when viewed from a left rear position at a certain angle.

(2) FIG. 2 is a cross-sectional view of the vehicle storage device shown in FIG. 1, viewed from the left.

(3) FIG. 3 is an exploded perspective view of a bracket body, a support portion, a shaft member, a lid and a coil spring shown in FIG. 2.

(4) FIG. 4 is an enlarged view of a part 4 in FIG. 2.

(5) FIG. 5A is a view when looked at in the direction of the arrow 5A in FIG. 3.

(6) FIG. 5B is a view when looked at in the direction of the arrow 5B in FIG. 3.

(7) FIG. 5C is a view when looked at in the direction of the arrow 5C in FIG. 3.

(8) FIG. 6A is a view useful to describe a temporary fixing step.

(9) FIG. 6B is a view useful to describe a lid closing step.

(10) FIG. 7 is a view useful to describe insertion of an engaging portion into an engaged portion.

(11) FIG. 8A is a view useful to describe relationship between the bracket body and the lid at the time of the temporary fixing step.

(12) FIG. 8B is a cross-sectional view taken along the line 8B-8B in FIG. 8A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) Embodiments of the present invention will be described below with reference to the accompanying drawings. In the following description, the terms “left” and “right” refer to the left and right when viewed from a person in a vehicle, and the terms “front” and “rear” refer to the front and rear when viewed in the traveling direction of the vehicle. In addition, “Fr” in the drawing represents the front, “Rr” represents the rear, “Le” represents the left when viewed from the person in the vehicle, “Ri” represents the right when viewed from the person in the vehicle, “Up” represents the top or up, and “Dn” represents the bottom or down.

Embodiments

(14) Referring to FIG. 1, a vehicle storage device **10** (hereinafter, referred to as “storage device **10**”) is, for example, a center console provided between a driver's seat and a passenger's seat in a vehicle cabin. In the following description, the center console will be described as an example of the storage device **10**.

(15) Referring FIGS. 2 and 3, the storage device **10** includes a storage portion **20** having an open top surface and capable of storing one or more goods of a person in the vehicle (hereinafter referred to as “storage items” or “things”), a bracket body **30** fixed to a rear surface portion **21** which is a rear surface of the storage portion **20**, support portions **40** and **40** extending rearward from left and right ends of the bracket body **30**, a shaft member **50** supported by the support portions **40** and **40**, a lid **70** swingably supported by the shaft member **50** and capable of opening and closing the opening of the storage portion **20**, a coil spring **80** provided to surround the shaft member **50** and urge the lid **70** in an opening direction, and a lid holding portion **17** provided in a front upper portion of the storage portion **20** and capable of holding the lid **70** in a closed state.

(16) The storage portion **20** is formed of resin and has a substantially rectangular parallelepiped shape with only an upper surface opened. The rear surface portion **21** may be referred to as an attaching surface portion **21** to which the bracket body **30** is attached. In the following description, the rear surface portion **21** may occasionally be referred to as an attaching surface portion **21**.

(17) Referring to FIG. 4, the storage portion **20** has positioning protrusions **21a** that protrude rearward from the attaching surface portion **21** to guide an attaching position of the bracket body **30**. Each of the positioning protrusions **21a** has a circular truncated cone shape whose diameter decreases toward the tip. Alternatively, each of the positioning protrusions **21a** may be a cross-shaped rib.

(18) The attaching surface portion **21** also has engaged portions **21b**, each of which protrudes rearward in a substantially U shape, such that a lower end of the bracket body **30** is inserted in the engaged portions **21b**. Those portions of the attaching surface portion **21** which face the engaged portions **21b** are referred to as bottom portions **21c**.

(19) Incidentally, since the only role of the respective engaged portion **21b** is to receive the bracket body **30**, the respective engaged portion **21b** may have a shape other than substantially U shape, such as substantially L shape. Further, as long as the engaged portion **21b** can receive the bracket body **30**, a lower surface of the engaged portion **21b** may be open. The engaged portion **21b** is required to have the open port to receive the bracket body **30**.

(20) Further, the bottom portion **21c** is not an indispensable element, and this portion may be dispensed with. It should be noted, however, that preferably the portion that faces the engaged portion **21b** is closed by the bottom portion **21c** rather than a hole. The reason will be described later.

(21) Referring also to FIG. 3, the bracket body **30** is constituted by a single metal plate together with the support portions **40** and **40**. The bracket body **30** includes a general surface portion **31** that defines a general face along the attaching surface portion **21** and has a substantially rectangular planar shape, a protruding surface **32** that protrudes toward the attaching surface portion **21** from the general surface portion **31**, a spring fixing portion **33** on which a part of the coil spring **80** is fixed, engaging portions **34** and **34** extending downward from the left and right lower ends of the general surface portion **31**, a rear extending portion **35** that bends rearward from the upper end of the general surface portion **31** and extends rearward, screw through holes **36** formed in the general surface portion **31** such that screws **18** fastened to the storage portion **20** extend through the holes **36** respectively, and positioning holes **37** formed in the general surface portion **31** such that the positioning protrusions **21a** extend through the positioning holes **37** respectively.

(22) The protruding surface **32** is formed in a lower center area of the general surface portion **31** and has a generally rectangular shape. A substantially rectangular abutment hole **32a** is formed in a center area of the protruding surface **32**.

(23) The spring fixing portion **33** is a tongue-shaped portion that extends upward, and is configured to allow the coil spring **80** to hook thereon.

(24) Each of the engaging portions **34** includes an abutting portion **34a** that can abut against the respective engaged portion **21b** and extends substantially parallel to the general surface portion **31**, and a connecting portion **34b** that connects the general surface portion **31** to the abutting portion **34a** and extends obliquely relative to the general surface portion **31** and the abutting portion **34a**.

(25) The screw through holes **36** are formed at four positions corresponding to the respective corners of the general surface portion **31** such that the four through holes **36** surround the protruding surface **32**.

(26) The inner diameter of each of the positioning holes **37** is slightly larger than the outer diameter of a root portion of the respective positioning protrusion **21a**. The two positioning holes **37** are formed on the right and the two positioning holes **37** are formed on the left. The screw through holes **36** are formed below the positioning holes **37**, respectively. The number and the positions of the positioning protrusions **21a** and the positioning holes **37** may be appropriately decided under given conditions. This is also true for the screws **18** and the screw through holes **36**.

(27) The support portions **40** and **40** extend downward respectively from the left and right ends of the rear extending portion **35**. A support portion through hole **41** is formed in each of the support portions **40** such that the shaft member **50** extends through the support portion through holes **41**.

(28) Referring to FIG. 5A, each of the support portion through holes **41** includes a circular hole portion **41a** that is formed in a substantially circular shape, and a protruding hole portion **41b** that protrudes radially outward from the circular hole portion **41a**.

(29) Referring to FIG. 3, the shaft member **50** includes a rod-shaped shaft portion **51** that extends through the left and right support portions **40** and **40** and the lid **70**, and collars **60** and **60** that surround the shaft portion **51** and provided in the respective support portion through holes **41**.

(30) Each of the collars **60** includes a shaft member insertion portion **61** that fits in the support portion through hole **41**, a shaft member flange portion **62** that is formed adjacent to the shaft member insertion portion **61** and has an outer diameter larger than the inner diameter of the support

portion through hole **41**, and a bearing portion **63** that extends from the shaft member flange portion **62** in the axial direction such that the coil spring **80** can contact the outer peripheral surface of the bearing portion **63**.

(31) The length of the shaft member insertion portion **61** in the axial direction is greater than the plate thickness of the respective support portion **40**. Therefore, the free end of the shaft member insertion portion **61** protrudes from the respective support portion **40**.

(32) Referring to FIGS. 5A and 5B, the outer peripheral surface of the shaft member insertion portion **61** has a shape similar to the inner peripheral surface of the support portion through hole **41**. The shaft member insertion portion **61** includes a circular insertion portion **61a** that is inserted into the circular hole portion **41a**, and a protrusion **61b** that protrudes radially outward from the circular insertion portion **61a** and is inserted into the protruding hole portion **41b**. As the protrusion **61b** is inserted into the protruding hole portion **41b**, the collar **60** is prevented from rotating.

(33) Referring to FIG. 3, the lid **70** includes left and right lid arm portions **71** and **71** that are respectively disposed on the left of the left support portion **40** and on the right of the right support portion **40** such that the shaft body **51** extends through the lid arm portions **71** and **71**, and a lid body **72** that is disposed in front of the lid arm portions **71** and **71** such that the lid body **72** can open and close the top of the storage portion **20** (see FIG. 2).

(34) Each of the lid arm portions **71** has a spring bearing portion **71a** that receives the biasing force of the coil spring **80**. The spring bearing portion **71a** is a concave portion that is formed in a substantially U shape along the biasing direction of the coil spring **80**.

(35) Referring also to FIG. 5C, a lid body protrusion **71b** is formed on that surface of each of the lid arm portions **71** which faces the support portion **40** such that the lid protrusion **71b** protrudes toward the support portion **40** and surrounds the circular insertion portion **61a** of the shaft member insertion portion **61**. The free end of the lid protrusion **71b** abuts against the protrusion **61b** when the lid **70** swings. As the free end of the lid protrusion **71b** abuts against the protrusion **61b**, the swinging of the lid **70** can be regulated. The free end of the lid protrusion **71b** may be referred to as a stopper portion **71c** that can abut against the protrusion **61b**.

(36) A lid through hole **71d** through which the shaft body **51** extends is formed in that portion of the lid arm portion **71** which is surrounded by the lid protrusion **71b**. The lid through hole **71d** is a round hole.

(37) Referring to FIGS. 2 and 3, a hook portion (reference numeral **73** in FIG. 6A) may be formed at the front end of the lid body **72** such that the hook portion hooks onto the lid holding portion **17**.

(38) Referring to FIG. 3, the coil spring **80** includes a left winding portion **81** and a right winding portion **81** such that the winding portions **81** are wound along the axial direction to impart a biasing force in the twisting direction. The coil spring **80** also includes one end fixing portion **82** that connects one end of one of the winding portions **81** to one end of the other winding portion **81** and is fixed to the spring fixing portion **33**, and distal end portions **83** and **83** that extend, in a substantially L shape, from the respective other ends of the winding portions **81** and **81** and bias the lid arm portions **71** and **71**.

(39) Referring to FIG. 2, the lid holding portion **17** may employ a well-known configuration. For example, the lid holding portion **17** may have an operation handle that is operable by a person in the vehicle, and a latch that is displaced upon operating the operation handle. As the person in the vehicle operates the operation handle to displace the latch, the locked state (engagement) between the hook portion of the lid **70** and the latch of the lid holding portion **17** is released. When the locked state is released, the front end of the lid **70** is caused to swing upward by the biasing force of the coil spring **80**. As a result, the upper portion of the storage portion **20** is opened. That is, the storage device **10** is brought into the open condition. When closing the lid **70**, a person in the vehicle causes the front end of the lid **70** to swing toward the storage portion **20** against the biasing force of the coil spring **80**. By squeezing the front end of the lid **70**, the lid **70** is engaged with the latch. The opening of the storage portion **20** is closed by the lid **70**. That is, the storage device **10** is

brought into the closed condition.

(40) Next, a method of assembling the storage device **10** will be described.

(41) Referring FIGS. **2** and **3**, firstly, the storage portion **20**, the bracket body **30**, the shaft member **50**, the lid **70** and the coil spring **80** are prepared (preparation step).

(42) Subsequently, the bracket body **30**, the shaft member **50**, the lid **70** and the coil spring **80** are temporarily assembled (temporary assembly step). More specifically, the bearing portions **63** and **63** are inserted into the winding portions **81** and **81**, and the shaft member insertion portions **61** and **61** are inserted into the support portion through holes **41** and **41**. Then, the support portion through holes **41** and **41** are aligned with the lid through holes **71d** and **71d**, and the shaft body **51** is inserted into the through holes **41**, **41**, **71d** and **71d**.

(43) Referring to FIGS. **6A** and **7**, next, the engaging portions **34** of the bracket body **30** are inserted into the engaged portions **21b** to temporarily fix the bracket body **30** to the storage portion **20** (temporary fixing step). At the time of temporary fixing, it is preferable that the positioning protrusions **21a** fit in the positioning holes **37** after the engaging portions **34** are inserted into the engaged portions **21b**.

(44) Referring to FIGS. **8A** and **8B**, when the temporary fixing is performed, the stopper portions **71c** of the lid **70** abut against the protrusions **61b** formed on the collars **60**. As a result, the bracket body **30** and the lid **70** can be kept substantially straight. The angle formed between the bracket body **30** and the lid **70** is preferably set to an angle that facilitates the work of temporary fixing, and can be set by the positions of the stopper portions **71c**.

(45) Referring to FIG. **6B**, next, it is preferable that the opening of the storage portion **20** is closed by the lid **70**, and the lid body **70** is held by the lid holding portion **17** (see FIG. **2**) provided at the storage portion **20** (lid closing step).

(46) Referring to FIG. **4**, next, the bracket body **30** that is temporarily fixed to the storage portion **20** is fixed to the attaching surface portion **21** (non-temporary fixing step). More specifically, the bracket body **30** is fixed to the attaching surface portion **21** with the screws **18**. This completes the assembly of the storage device **10**. Incidentally, the non-temporary fixing may be carried out by means of certain elements other than screwing.

(47) The above-described storage device **10** will be summarized below.

(48) Referring to FIGS. **2** and **4**, the storage device **10** includes the storage portion **20** that has one open face and can store thing(s) to be stored therein, the bracket body **30** fixed to the attaching surface portion **21**, which is one face of the storage portion **20**, the shaft member **50** supported by the bracket body **30**, the lid **70** that is swingably supported by the shaft member **50** such that the lid **70** can open and close at least part of the opening of the storage portion **20**, and the coil spring **80** that biases the lid **70** in the opening direction. The bracket body **30** includes the engaging portions **34** extending along the attaching surface portion **21**. The attaching surface portion **21** has the engaged portions **21b** that define the openings into which the engaging portions **34** are inserted.

(49) The assembling of the storage device **10** is performed through the preparation step of preparing the storage portion **20** having the single open face and the engagement portions **21b** formed on the attaching surface portion **21**, the bracket body **30** having the engaging portions **34**, the shaft member **50**, the lid **70**, and the coil spring **80**, the temporary assembly step of temporarily assembling the bracket body **30**, the shaft member **50**, the lid **70**, and the coil spring **80**, the temporary fixing step of inserting the engaging portions **34** of the bracket body **30**, on which the lid **70** and the coil spring **80** are temporarily assembled, into the engaged portions **21b** and temporarily fixing the bracket body **30** to the storage portion **20**, and the fixing step of fixing the bracket body **30**, which is temporarily fixed to the storage portion **20**, to the attaching surface portion **21**.

(50) The attaching of the bracket body **30** to the attaching surface portion **21** is performed while the engaging portions **34** are being inserted in the engaged portions **21b**. This eliminates the need to support the weight of the lid **70** when performing the attaching work. Further, there is no need to support the bracket body **30**. As a result, the storage device **10** can easily be assembled.

(51) It should be noted that it is preferable to further include, between the temporary fixing step and the fixing step, a lid closing step of closing at least a part of the opening of the storage portion **20** with the lid **70** such that the lid holding portion **17** provided on the storage portion **20** holds the lid **70**.

(52) By keeping the lid **70** in the closed condition, wobbling or the like of the lid **70** can be suppressed or avoided. Further, the bracket body **30** is not caused to swing by the biasing force of the coil spring **80**, and therefore the force that presses toward the attaching surface portion **21** is not required. Thus, the attaching of the bracket body **30** to the attaching surface portion **21** is further facilitated. Consequently, the assembly of the storage device **10** is more facilitated.

(53) Referring to FIG. 7, the bottom portions **21c** are formed on the storage portion **20** such that the bottom portions **21c** face the respective engaged portions **21b** and the free ends of the engaging portions **34** can contact the bottom portions **21c** at the time of inserting the engaging portions **34**. The bottom portions **21c** function as guides, and therefore it is possible to more easily perform the insertion of the bracket body **30** into the engaged portions **21b**. Thus, the assembly of the storage device **10** is more facilitated.

(54) Furthermore, the lower part of the respective engaged portion **21b** is closed by a lower surface portion **21d**. This makes it possible to temporarily assemble the bracket body **30** to a more appropriate position in an easy manner. Incidentally, the lower surface portion **21d** may be referred to as an opening facing portion **21d** that closes the position facing the opening of the respective engaged portion **21b**.

(55) Further, each of the engaging portions **34** includes the abutting portion **34a** that can abut against the respective engaged portion **21b** and extends substantially parallel to the general surface portion **31**, which constitutes the general surface of the bracket body **30**, and the connecting portion **34b** that connects the general surface portion **31** to the abutting portion **34a** and extends obliquely relative to the general surface portion **31** and the abutting portion **34a**. As a result, the abutting portion **34a** can be formed at a position closer to the attaching surface portion **21** than to the general surface portion **31**. Compared with a configuration in which the general surface portion **31** and the abutting portion **34a** are present on the same plane, insertion of the engaging portions **34** into the engaged portions **21b** is facilitated. Thus, the assembly of the storage device **10** is more facilitated.

(56) Referring to FIGS. 6A and 8B, the shaft member **50** has the protrusions **61b** protruding radially outward. In addition, the lid **70** has the stopper portions **71c** that can abut on the protrusions **61b** respectively when the lid **70** reaches a predetermined opening angle with respect to the bracket body **30**. Thus, it is possible to keep the bracket body **30** and the lid **70** at predetermined opening angles (or predetermined opening degrees) respectively during the assembly work. By keeping them at the predetermined opening angles, it is possible to suppress the wobbling or the like of the bracket body **30** during the attaching work. Thus, the assembly of the storage device **10** is more facilitated.

(57) The storage device of the present invention is suitable for a center console of a passenger vehicle. It should be noted, however, that the storage device of the present invention is also applicable to a glove compartment or the like. Further, the present invention is not limited to those configurations that insert the engaging portions from above the engaged portions, i.e., the present invention also encompasses those configurations that insert the engaging portions into the engaged portions in a horizontal direction or obliquely relative to the vertical direction. In other words, as long as the operations and advantages of the present invention are achieved, the present invention is not limited to the embodiments.

Claims

1. A storage device for a vehicle, comprising: a storage portion that has an opening in its one face and is configured to store items therein; a bracket body fixed to an attaching surface portion, which is another face of the storage portion; a shaft member supported on the bracket body; a lid supported on the shaft member such that the lid can swing to open and close at least part of the opening of the storage portion; and a coil spring configured to bias the lid in an opening direction, the bracket body including at least one engaging portion that extends along the attaching surface portion, wherein the at least one engaging portion is integrally provided by the bracket body and extends therefrom; and the attaching surface portion including at least one engaged portion that has an opening to receive the at least one engaging portion, wherein the storage portion has at least one bottom portion that faces the at least one engaged portion respectively such that a free end of each of the at least one engaging portion can abut against the at least one bottom portion when the at least one engaging portion is inserted into the at least one engaged portion.
 2. The storage device for the vehicle according to claim 1, wherein each of the at least one engaging portion has an abutting portion that can abut against the respective engaged portion and extends substantially parallel to a general surface portion, which constitutes a general surface of the bracket body, and a connecting portion that connects the general surface portion to the abutting portion and extends obliquely relative to the general surface portion and the abutting portion.
 3. A storage device for a vehicle, comprising: a storage portion that has an opening in its one face and is configured to store items therein; a bracket body fixed to an attaching surface portion, which is another face of the storage portion; a shaft member supported on the bracket body: a lid supported on the shaft member such that the lid can swing to open and close at least part of the opening of the storage portion; and a coil spring configured to bias the lid in an opening direction, the bracket body including at least one engaging portion that extends along the attaching surface portion, wherein the at least one engaging portion is integrally provided by the bracket body and extends therefrom; and the attaching surface portion including at least one engaged portion that has an opening to receive the at least one engaging portion, wherein the shaft member has at least one protrusion that protrudes radially outward, and the lid has at least one stopper that can abut against the at least one protrusion respectively when the at least one stopper reaches a predetermined opening angle relative to the bracket body.
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